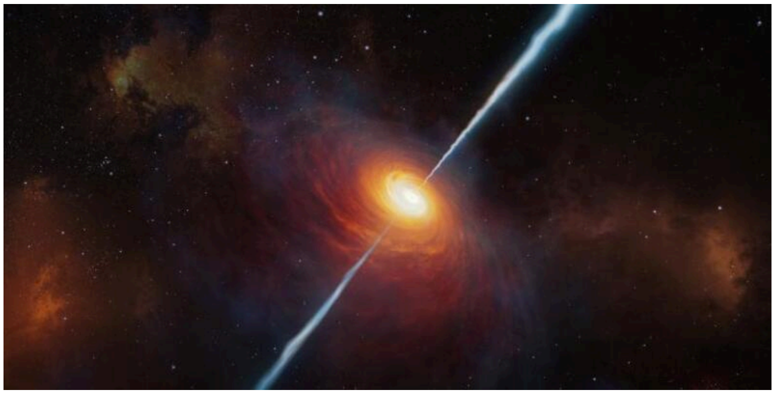
Assignment 3 – Night Sky Star Classification using Neural Networks

Problem Statement



The problem entails the development of a machine learning model for Night Sky Classification, specifically targeting the detection of quasars from celestial images captured by telescopes. Quasars, or quasi-stellar radio sources, are luminous and distant astronomical objects that emit large amounts of energy. Accurately identifying quasars amidst various celestial objects in the night sky is pivotal for advancing our understanding of the universe's structure, evolution, and cosmology.

Importance:

Understanding quasars is crucial in several aspects of astrophysics and cosmology. Quasars are believed to be powered by supermassive black holes at the centers of galaxies, making them natural laboratories for studying extreme physical phenomena. By analyzing quasar spectra and variability, scientists can probe the properties of black holes, galaxy formation, and the intergalactic medium. Quasars also serve as cosmological beacons, providing valuable information about the large-scale structure of the universe, including its expansion rate and the distribution of matter.

Accurate classification of quasars from observational data is vital for numerous astronomical endeavors, including:

Cosmological Studies: Quasars serve as standard candles for measuring cosmic distances, enabling precise measurements of the universe's expansion rate and the evolution of large-scale structures.

Galaxy Evolution: Quasars are intimately linked to the formation and evolution of galaxies. Studying the environments and properties of quasar host galaxies sheds light on the coevolution of supermassive black holes and their host galaxies over cosmic time.

Dark Matter and Dark Energy: Quasar surveys can probe the distribution of dark matter and dark energy in the universe, providing insights into their nature and influence on cosmic structures.

Early Universe: Quasars are among the most distant objects observed in the universe, offering glimpses into the early stages of cosmic evolution shortly after the Big Bang.

The Dataset

The “Stellar Classification Dataset — SDSS17” by FEDESORIANO is made from the “Sloan Digital Sky Survey” project and contains spectroscopic observations of celestial objects in the night sky. The dataset consists of 100,000 observations, each described by 17 feature columns and 1 class column that identifies it as a star, galaxy, or quasar.

Quasars, short for “quasi-stellar radio sources,” are incredibly bright and distant astronomical objects found at the centers of galaxies. Quasars are not stars but the active cores of distant galaxies powered by supermassive black holes.

Data Dictionary

1. obj_ID = Object Identifier, the unique value that identifies the object in the image catalog used by the CAS (Target variable)
2. alpha = Right Ascension angle (at J2000 epoch)
3. delta = Declination angle (at J2000 epoch)
4. u = Ultraviolet filter in the photometric system
5. g = Green filter in the photometric system
6. r = Red filter in the photometric system
7. i = Near Infrared filter in the photometric system
8. z = Infrared filter in the photometric system
9. run_ID = Run Number used to identify the specific scan
10. rereun_ID = Rerun Number to specify how the image was processed
11. cam_col = Camera column to identify the scanline within the run
12. field_ID = Field number to identify each field
13. spec_obj_ID = Unique ID used for optical spectroscopic objects (this means that 2 different observations with the same spec_obj_ID must share the output class)
14. class = object class (galaxy, star or quasar object)
15. redshift = redshift value based on the increase in wavelength
16. plate = plate ID, identifies each plate in SDSS
17. MJD = Modified Julian Date, used to indicate when a given piece of SDSS data was taken
18. fiber_ID = fiber ID that identifies the fiber that pointed the light at the focal plane in each observation

Instructions

Load the data to answer the following questions:

1. Data information – List out the how many numerical and categorical variables
2. Perform feature statistics and create a table for the numerical variables' mean mode, dispersion, min, and max. Write down observations for all the numerical variables.
3. Perform detailed EDA to analyze a data set
4. Build a classifier workflow with all features using different neural network architectures
5. Explain in your detail your machine learning approach
6. Perform output analysis for your generalized machine learning model you are deploying
7. Your slide deck should be 10 -12 slides minimum and should include a detailed final summary

Submission

Along with PowerPoint, please upload a copy of*.ows file
Naming convention – StudentNameAssignment_#.ppt, StudentNameAssignment_#.ows